Adaptive Workspaces: Evaluating AI-Proposed Interface Elements with Interaction Logs, Eye Tracking, and EEG

Milestone 2 — Prototype + Method / Evaluation Plan

Team: Julian Mariscal · Gary DeVilbiss · Joshua Joubert — Worcester Polytechnic Institute · CS 546 · Prof. Erin Solovey **Track:** Artifact Contribution (interactive system + study design) **Prototype:** `adaptive-workspaces` — local-only web app ([repo / demo link here]) **Demo:** runs fully offline on one laptop; seeded click-through at `/demo`

> **Fill-in before submitting:** the repo/demo link, the per-person Milestone 2 contributions (last section — confirm the split), and the exact GenAI tools you used in your process. Citation numbers `[n]` refer to your Milestone 1 reference list.

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1. One-Page Status Document

What we built. A working research probe: a browser spreadsheet in which every user action is captured as a structured command. A deterministic *repeat miner* watches the command stream for sequences the user repeats across sheets; when a repeat crosses a support threshold *K*, a **local** LLM (qwen3:8b via Ollama, running offline on the same laptop) names the repeated routine and writes a one-sentence proposal card. The user accepts, edits, or dismisses the card; on accept, a one-click tool appears on a "My Tools" toolbar and re-applies the routine to any sheet — re-targeting columns **by header**, so it works even when a later sheet has its columns in a different order. The app supports three interaction conditions (default / prompted / proactive), per-stream informed consent, NASA-TLX after each block, and lab-grade sensing (eye-tracking + EEG) over Lab Streaming Layer (LSL). Every event — including when a proposal appeared and whether it was accepted, dismissed, or later removed — is written to a local JSONL log. **The central mechanism (proactive mine → LLM proposal → accept → one-click reuse) is functional and verified end-to-end** by an automated browser test against the live LLM, plus 77 unit tests.

What we learned (from a pilot run on our own task, see §4). (1) The assistant only fires after a routine repeats across *two* sheets; in an early run the block was ended after a single sheet and the assistant never appeared — so a correct system looked broken. (2) The manual toolbar's one-click "do everything" buttons undersold the tedium the assistant is meant to relieve. (3) Sorting a cleaned sheet scrambled the total row. Each finding drove a concrete change (§4).

What we will change before Milestone 3. Recruit and run the full evaluation (≥7 participants for M3) across the three conditions with the lab sensing stack; finalize the exit-interview protocol; and analyze the eye-tracking + EEG streams post-session against the proposal-event markers. Open risk we are managing: lab-sensing throughput — booking and calibrating eye-tracker + EEG for every participant by early December.

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2. Method / Evaluation Plan

Research questions (from Milestone 1)

- **RQ1 (Proactive vs. prompted).** Do AI-proposed elements, derived from observed repeated actions, reduce task time and steps compared with elements users request by prompt and with the default interface?
- **RQ2 (Adoption).** Which proposed elements do users accept, keep using, or remove, and why?
- **RQ3 (Hidden costs).** Do eye-tracking and EEG signals reveal costs of adaptation that decrease work efficiency?

These map directly onto the three study conditions: **default** (baseline), **prompted** (user requests an element, in the spirit of DynaVis [6]), and **proactive** (the system mines repeated actions and proposes an element unprompted) — the comparison at the heart of RQ1.

2a. Study Design

Design. Within-subjects, single session (~45 min). Each participant completes the same spreadsheet-cleanup task under **three conditions**, one per block:

- **A — Default:** the toolbar only; no assistant panel.
- **B — Prompted:** an "Ask for a tool" box; the participant describes a tool in natural language and the system builds it (user-initiated automation).
- **C — Proactive:** the assistant watches for repeats and offers a proposal card unprompted (system-initiated automation).

Counterbalancing. Condition order and workbook assignment are assigned from the participant ID via a Latin square (implemented in the prototype), so practice/fatigue effects do not confound the condition comparison and no participant sees the same data twice. Three **equivalent** workbooks (A/B/C) are generated from fixed seeds: same structure and difficulty, different values; two of the six sheets in each workbook have shuffled column order so the "re-target by header" behavior is exercised in every condition.

Task & stimuli. Each workbook is six "monthly export" sheets (Jan–Jun) of synthetic sales rows. For every sheet the participant performs the same cleanup: delete the blank rows (one at a time, selecting each), bold and freeze the header, format Amount as currency and Date as a date, and add a SUM total row under Amount and Qty. The routine is deliberately repetitive across the six sheets — that repetition is the phenomenon the assistant targets.

Procedure. (1) Consent, with a separate checkbox per data stream (interaction log / eye-tracking / EEG). (2) Sensing calibration (eye-tracker; EEG signal check). (3) A 2-minute unscored practice sheet. (4) Three task blocks (one per condition), each followed by a NASA-TLX questionnaire. (5) A short semi-structured exit interview. Five LSL sync pulses are emitted at session start so the interaction log, eye-tracking, and EEG streams can be aligned offline.

Pilot scope (this milestone). We piloted the two most contrasting conditions — **Default (A)** and **Proactive (C)** — which is the comparison that most directly tests the core claim; the full study adds the Prompted (B) condition. (See §4 for pilot results.)

Ethics (consent, data handling, debrief). Participation is voluntary with the right to withdraw at any time. Consent is collected per data stream, so a participant may allow interaction logging but decline eye-tracking or EEG. **All data stays on the local machine** (`data/sessions/<pid>/<session>.jsonl`): no cloud APIs, no telemetry, no analytics at runtime. The workbooks are **synthetic** — participants never handle real or personal data. The on-device LLM sees only the *abstract operation sequence* (e.g., "delete empty rows → bold header → ...") and generic synthetic column headers; **no cell values and nothing identifying ever leave the laptop**, and the model is not trained or fine-tuned on participant data. This directly operationalizes what our Milestone 1 interviewees asked for (data should stay local and should not train the AI). Participants are debriefed on the study's purpose at the end and may have their session deleted on request.

2b. Evaluation / Measures

The prototype already logs every measure below with aligned timestamps.

Measure	Source in the log	RQ
Task time per sheet	command timestamps (first→last command per sheet; block_start→block_end)	RQ1
Manual action count per sheet	`command` events, `source: manual`	RQ1
Clicks saved by tools	`tool_used` events × bundled step count	RQ1
Accept / dismiss / "don't suggest" / remove rates	`assistant` events (`accepted`, `not_now`, `never`, `tool_removed`)	RQ2
Time-to-accept	latency `proposal_shown` → `accepted`	RQ2
Attention: gaze on spreadsheet vs. toolbar vs. assistant panel	gaze stream × logged AOI rectangles (`aw-grid`, `aw-toolbar`, `aw-mytools`, `aw-assistant`)	RQ3
EEG workload	Unicorn EEG, analyzed **post-session** from the CSV export, windowed around proposal-event markers	RQ3
Subjective workload	NASA-TLX, six subscales, after each block	RQ1, RQ3
Trust / interruption / control	semi-structured exit interview (coded for themes)	RQ2

Primary analysis is the within-subjects contrast across the three conditions (A vs. B vs. C) on task time, action count, and NASA-TLX (RQ1), with proposal-response rates (RQ2) and gaze/EEG (RQ3) as the mechanism-level measures explaining *why* a difference arises. The sensing analysis is grounded in our Milestone 1 reading: EEG stress markers (decreased alpha, increased beta power [7]) and the use of combined EEG + eye-tracking features to tell apart target from non-target fixations [8], following Liu et al.'s multimodal approach of aligning physiological signals to task

events [4]. We use these signals, as the abstract states, *to evaluate* whether a generated element eases work or adds cognitive load — not to trigger the AI. Subjective workload uses NASA-TLX (add the Hart & Staveland 1988 citation to your reference list if you report it); order effects are controlled by Latin-square counterbalancing.

2c. Ecological Validity

The task is a near-direct instantiation of what **P2** (Excel / Power BI) described in our Milestone 1 interviews: *"resetting column sizes (A, B, C, D) every time he makes another table on another sheet — you're constantly going back and forth,"* plus reformatting cells, adding headings, and rebuilding totals on *"new workbooks over and over … time wasted."* Our six-sheet monthly-export cleanup reproduces exactly that pattern — the same multi-step routine repeated across structurally-similar sheets — which is precisely the condition under which a proactive assistant could help. The stimulus therefore reflects the phenomenon our users actually named rather than an artificial puzzle. We strengthened this during iteration (§4): the manual toolbar now requires the participant to *aim* each operation (select the target, then act) and to delete blank rows one at a time — as real spreadsheet work does — instead of one magic button that cleans the whole sheet. This makes the baseline's effort realistic and keeps row-deletion a human judgment, matching the interview finding that users want to review destructive actions themselves.

What is still artificial, and we say so. Data and workbooks are synthetic; the task is fixed rather than self-chosen; a single lab session compresses what is really a recurring, multi-session habit; and the lab sensing setup is not a natural work environment. We frame the single-session, six-sheet design as a tractable proxy and describe the multi-session, in-the-wild version as the ideal study.

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3. Built Work (Prototype)

- **Repo / files:** [link]
- **Run it:** `npm install && ollama pull qwen3:8b && npm run dev`, then open `/demo` for a

seeded click-through or `/setup` for the researcher screen. Everything runs offline.

- **Verification:** an automated end-to-end browser test drives the full proactive loop

(mine → LLM proposal → accept → one-click reuse on a shuffled-column sheet) against the live local LLM and checks the resulting sheet and event log; 77 unit tests cover the command engine, miner, generalizer, and server. All passing.

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4. Iteration (pilot findings → changes)

We piloted the Default and Proactive conditions on our own task and instrumented the session. Three issues surfaced; each produced a change, with before/after artifacts.

1. **The assistant needs two sheets, and that wasn't obvious.** In a pilot run the block was

ended after cleaning a single sheet, so the repeat never occurred and the assistant never proposed anything — a correct system that *looked* broken. **Change:** a live progress nudge in the assistant panel that tells the participant exactly what unlocks a suggestion ("you've cleaned Jan; now clean another sheet the same way"). *Artifacts: pilot log* `DEMO_demomw9lfwcn.jsonl`*; before/after screenshots of the assistant panel.*

1. **The manual baseline was too easy, undercutting the premise.** One-click buttons like

"Delete empty rows" did the whole job, so the repetition the assistant is meant to relieve didn't *feel* tedious. **Change:** every toolbar operation now requires aiming first (select, then act), and blank rows are deleted one at a time — closer to real spreadsheet work and to our interviewees' desire to stay in control of deletions. *Artifacts: before/after toolbar screenshots;* `DECISIONS.md`.

1. **Sorting broke the total row.** Sorting a cleaned sheet dragged the SUM total row up into

the data. **Change:** total rows are now pinned at the bottom during a sort. *Artifacts: failing-vs-fixed screenshots; regression test added.*

Our pilot NASA-TLX also showed the expected direction — far higher effort and frustration in the manual Default block than when the assistant was available — which is encouraging evidence that the manipulation has the intended effect, to be confirmed with real participants at Milestone 3.

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5. AI Module 1 — Where the LLM Got Our Users Wrong

Validating a naive "let the LLM watch and automate" design against our Milestone 1 interview data surfaced four places the AI would have gotten our users wrong — each became a design decision:

- **"Notify first," not act silently.** P1 was explicit: *automatic* changes should notify or

ask the user before executing, while *prompted* ones can run right away. A proactive LLM that just applies changes violates this. We reconciled it by making every proactive change a *proposal card that notifies but never auto-applies* (accept / edit / "not now" / "don't suggest this"), one card at a time, never stealing focus — while the prompted condition runs immediately after the user's request, exactly the asymmetry P1 described.

- **Repetition threshold, which the LLM alone doesn't know.** P2 would welcome an auto-added

tool *only if the pattern is caught within 2–3 sessions*; otherwise he'd rather prompt for it himself. An LLM asked to "suggest automations" has no notion of how many repeats justify a suggestion. Our **deterministic miner gates proposals on a support threshold K** (and the prompted condition exists precisely for patterns below it), operationalizing P2's "2–3 sessions" rule rather than trusting the model's judgment about when to speak up.

- **Over-reaching into destructive, data-dependent actions.** The model will happily bundle

everything a user repeats, including blank-row deletion — but our users wanted to keep control and be able to undo. We **exclude row deletion from the learned tool** (it also can't generalize — blanks sit in different places on every sheet) and keep it a manual, human-judged step. The assistant automates the deterministic formatting; the human decides what to delete.

- **Locality and trust.** P1 would only accept this if the AI were *local or enterprise and

isolated to one program*, and both interviewees were uneasy about data leaving the machine (and about EEG). A cloud LLM would fail that test outright. Our proposer is a **local model running offline**; it sees only abstract operation names and synthetic headers, and per- stream consent lets a participant decline eye-tracking or EEG entirely.

Finally, the model's role is deliberately narrow: it *names* a routine the miner already found — it never chooses the steps — and every label is validated, retried once, and falls back to deterministic text if it misbehaves. An LLM mistake can produce a bad label, never a broken spreadsheet.

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6. Team Contributions

(1–3 sentences each — confirm/adjust to your actual split before submitting.)

- **Julian Mariscal:** Built the AI/assistant layer — the repeat miner, the local-LLM proposer

with validation and deterministic fallback, and the proactive proposal UI — and led the pilot run and the prototype iteration (§4).

- **Gary DeVilbiss:** Led the sensing and evaluation plan — the eye-tracking + EEG measures and

post-session analysis approach (grounded in [4], [7], [8]) — and the ethics / consent and participant-recruitment plan. [confirm]

- **Joshua Joubert:** Led the study design and stimuli — the three-condition structure, task and

equivalent workbooks, and counterbalancing — and the measures-to-RQ mapping. [confirm]

7. Generative-AI Disclosure

Per the course Generative AI policy, we disclose our AI use:

- **In the system (studied, not incidental):** a local qwen3:8b model (Ollama, offline)

names mined routines and writes proposal-card text. This is the research artifact itself; it runs on-device and sees no participant data (§2a).

- **In our process:** we used [ChatGPT / Claude / etc.] for [ideation / code scaffolding /

drafting and editing this document / debugging]. All AI output was reviewed and validated by the team against our own code and our Milestone 1 user data; we take responsibility for the final content. [Edit this to match exactly what you used and where.]

Appendix — Process artifacts included

- `DECISIONS.md` — running design-decision log
- Pilot session log: `DEMO_demomw9lfwcn.jsonl`
- Before/after screenshots: assistant nudge; aim-first toolbar; sort/total fix
- Automated end-to-end test + unit-test run output